METRIC

MIL-DTL-82655A (NAVY)
20 April 2004
SUPERSEDING
MIL-F-82655
31 January 1977

DETAIL SPECIFICATION

FERRIC OXIDE, TECHNICAL

This specification is approved for use within the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for three classes of ferric oxide.

1.2 Classification. The ferric oxide will be of the following classes as specified in the contract or order (see 6.2):

Class 1
Class 2
Class 3

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents. This section is not applicable to this specification.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DOD adopted, are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 6810

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

ASTM INTERNATIONAL

ASTM D 3663 Standard Test Method for Surface Area of Catalysts and Catalyst Carriers

(Copies of this document are available online at http://www.astm.org/ or from ASTM International Customer Service, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be ferric oxide in powder form.

3.2 Chemical and physical properties. The chemical and physical properties shall be in accordance with Table I.

TABLE I. Chemical and physical properties.

Property	Class 1		Class 2		Class 3	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Moisture, wt%	–	0.5	–	0.5	–	0.5
Calcination loss, wt% (dry basis)	–	2.0	–	2.0	–	2.0
Iron, as Fe_2O_3, wt% (calcined basis)	98.0	–	98.0	–	98.0	–
Acidity, as H_2SO_4, wt%	–	0.02	–	0.02	–	0.02
Specific Surface Area, m^2/g	6.0	8.0	5.0	6.0	8.0	10.0
Typical Particle Shape	Acicular		Spheroid		Spheroid	

3.3 Workmanship. The material shall be uniform, free from contamination, foreign material, or any other defect that would prevent its use for the purpose intended.

4. VERIFICATION.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. Conformance inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

d. Humidity: Room ambient to 95 percent relative, maximum.

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.3.2 Sampling. Sampling for quality conformance inspection shall be no less than one unit of material. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests of 4.5 or shall meet the requirements of 4.4.1. When specified in the contract or order (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

4.4.1 Alternate test requirements. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests of 4.5. If the solid propellant produced using the material meets the requirements of the applicable propellant specification, the limits for specific surface area in Table 1 may be considered advisory and not mandatory.

4.5 Test methods. Unless an alternative inspection method is proposed, tests shall be performed using the apparatus and procedures specified herein. Alternative inspection methods may be proposed by the supplier but must be approved by the contracting activity prior to use in acceptance testing. Unless otherwise specified herein, all chemical reagents used for these tests shall be ACS grade or better.

4.5.1 Moisture content. The moisture content shall be determined in accordance with the following:

a. Transfer a sample of approximately 2 grams (g) to a tared 30 milliliter (ml) silica crucible. Cover the crucible with a tight-fitting tared cover and weigh to the nearest 0.1 mg.

b. Remove the cover and place the crucible in an oven at 105 to 110°C for 1 hour, minimum.

c. Replace the cover, remove the crucible from the oven, cool in a desiccator and weigh to the nearest 0.1 mg. Retain covered sample for use in the calcination loss test of 4.5.2.

d. Calculate the weight percent (wt%) moisture as follows:

$$\text{Moisture, wt\%} = \left(\frac{A-B}{A-C}\right) \times 100$$

where: A = Weight of crucible, cover and sample, g
B = Weight of crucible, cover and dried sample, g
C = Weight of crucible and cover, g

e. Report the results of 2 determinations and their average.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

4.5.2 Calcination loss. The calcination loss shall be determined on a dry basis in accordance with the following:

a. Remove the cover from the crucible with sample from 4.5.1c and place the crucible in a muffle furnace at 800 to 900°C for 30 minutes, minimum.

b. Remove the crucible from the furnace and allow to cool for 2 to 3 minutes. Replace the cover, place the crucible in a desiccator and cool to room temperature. Weigh the covered crucible to the nearest 0.1 mg. Retain covered sample for use in the iron content test of 4.5.3.

c. Calculate the calcination loss on a dry basis as follows:

$$\text{Calcination loss, wt\%} = \left(\frac{B-C}{B-D}\right) \times 100$$

where: B = Weight of crucible, cover and dried sample (from 4.5.1e), g
C = Weight of crucible, cover and ignited sample, g
D = Weight of crucible and cover (from 4.5.1e), g

d. Report the results of 2 determinations and their average.

4.5.3 Iron content. The iron content as weight percent Fe_2O_3 shall be determined on a calcined basis in accordance with the following:

a. Prepare or obtain the following reagents:
1. Hydrochloric acid, concentrated
2. Stannous chloride solution - prepare fresh by dissolving 12.5 g, weighed to the nearest 0.1 mg, of stannous chloride in 25 ml hot concentrated hydrochloric acid and diluting to 250 ml with distilled water.
3. Mercuric chloride, saturated solution - dissolve 100 g of mercuric chloride in 1 liter of distilled water by heating.
4. Sulfuric-phosphoric acid mixture - made by adding 150 ml concentrated sulfuric acid to 500 ml distilled water, adding 150 ml of 85 percent phosphoric acid and diluting to 1 liter with distilled water.
5. Barium diphenylamine sulfonate indicator, 0.2 percent.
6. Standard potassium dichromate solution, 0.1 normal (N).

b. Weigh 0.28 to 0.32 g of ignited sample from 4.5.2b to the nearest 0.1 mg into a 400 ml beaker.
.

c. Add 20 ml of concentrated hydrochloric acid and cover the beaker. Heat on a hot plate until the sample is completely dissolved. Remove the cover and rinse the cover and the sides of the beaker with 15 to 20 ml of distilled water.

d. With the beaker still on the hot plate, slowly add stannous chloride solution dropwise until the last drop makes the solution colorless or free from any tinge of yellow, then add an additional 1 or 2 drops. Remove the beaker from the heat and rapidly cool to room temperature. Quickly add 10 ml of mercuric chloride solution, swirl a few seconds until uniform in appearance, then set aside for 2 or 3 minutes to allow a slight white precipitate to form. If precipitate does not form or if a heavy grayish precipitate forms, discard and perform another determination using more or less stannous chloride, as indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

e. If precipitate forms correctly, add 200 ml of water, 20 ml of sulfuric-phosphoric acid mixture, and 6 to 8 drops of 0.2% barium diphenylamine sulfonate indicator. Titrate with standard potassium dichromate solution, taking the sudden change from gray-green to blue as the endpoint. Run a blank determination with the same reagents.

f. Calculate the wt% Fe_2O_3 on a calcined basis as follows:

$$\text{Iron as } Fe_2O_3, \text{ wt\%} = \left(0.07985 \frac{(V-v)N}{W} \right) \times 100$$

where: V = Volume of potassium dichromate solution used to titrate sample, ml
v = Volume of potassium dichromate solution used to titrate blank, ml
N = Normality of potassium dichromate solution
W = Weight of ignited sample used, g
0.07985 = Milliequivalent weight of Fe_2O_3

g. Report the results of 2 determinations and their average.

4.5.4 Acidity. The acidity as wt% H_2SO_4 shall be determined in accordance with the following:

a. Weigh 10 ± 0.5 g of sample to the nearest 0.1 mg. Add 100 ml of distilled water and boil for several minutes.

b. Filter and rinse sample with hot distilled water, adding rinsings to the filtrate.

c. Titrate the filtrate with 0.1N potassium hydroxide using methyl orange indicator.

d. Run a blank determination.

e. Calculate the acidity as wt% H_2SO_4 as follows:

$$\text{Acidity as } H_2SO_4, \text{ wt\%} = \left(\frac{0.04904 (V-v)N}{W} \right) \times 100$$

where: V = Volume of potassium hydroxide solution used to titrate sample, ml
v = Volume of potassium hydroxide solution used to titrate blank, ml
N = Normality of potassium hydroxide solution
W = Weight of sample, g
0.04904 = Milliequivalent weight of sulfuric acid

f. Report the results of a minimum of 2 determinations and their average.

4.5.5 Specific Surface Area. The specific surface area shall be determined by multipoint B.E.T. nitrogen adsorption in accordance with the methods outlined in ASTM Method D 3663 for Type II nitrogen adsorption isotherms. Material shall be dried from 4.5.1. Instrumental conditions may vary between equivalent analyzers and should be set in accordance with the manufacturer's instructions. Report the B.E.T. surface area expressed in square meters per gram.

4.5.6 Particle Size. The particle size distribution shall be determined for information only. Measurement of particle size distribution by a sedimentation method is preferred but not required. Report shall contain, at a

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

minimum, the following data: test instrument name, the mean particle size, and particle sizes expressed in microns at 10%, 50%, and 90% from a volume to particle size distribution curve.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory, unless otherwise stated in the contract or purchase order.)

6.1 Intended use. Ferric oxide in accordance with this specification is intended for use as an ingredient of solid propellant. Class 1 ferric oxide is specifically intended for use in the CCU-22/A impulse cartridge assembly. Class 2 ferric oxide is specifically intended for use in the CCU-149/A and CCU-22B/A impulse cartridge assemblies, and the CKU-5C/A rocket catapult assembly. Class 3 ferric oxide has been used in the CKU-5B/A rocket catapult assembly. Since these classes of ferric oxide were developed solely for use in military items, there are no commercial applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Quantity required
c. Place of delivery
d. Inspection conditions when other than as specified (see 4.2)
e. Lot size if other than as specified (see 4.3.1)
f. Packaging requirements if other than as specified (see 5.1)
g. Size of container required
h. Any special markings required

6.2.2 Data requirements. The following data requirements should be considered when this specification is applied on a contract. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Ref. Paragraph	Title
4.4	Quality conformance inspection data

6.3 Suggested source of supply. Products that have met the requirements of this specification in past procurement actions are RY2096, R5098R, and R2899R for Class 1, Class 2, and Class 3 respectively, manufactured by Elementis Pigments, 2051 Lynch Avenue, East St. Louis, IL 62204. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

CCU-22/A
CCU-22B/A
CCU-149/A
CKU-5C/A

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project 6810-1721)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.